SE 305-321
SOFTWARE SPECIFICATION AND DESIGN

Project Information

WEEK #4

Contact Information

- My e-mail address: cekirdekkhamza@gmail.com

- Please send me your questions via e-mail. Try not to use OASIS messages.

- We will use Blackboard for the announcements, documents, LAB content, and etc.

Project Information

- In this project, you are going to design a software application as a team.
- Your design should include various diagrams discussed in class.
- You should create your design using a software design application and prepare necessary reports throughout the semester.

Project Information (cont'd)

- At the end of the semester, you will be able to:
  - collect requirements from clients,
  - design a software system,
  - report a software design,
  - discuss design details from the perspective of software engineering discipline.

Project Information (cont'd)

- You need to form a group of 4 or 5 people from your section only.
- You are expected to create an object-oriented design according to the topics discussed in class.
- You need to find a real client.
  - Should be someone you can reach out and have a client meeting either online or at the client's site.
- Project topic should be selected by you and your team, and it should be a software related system.

Project Information (cont'd)

- You will NOT code anything.

- You are going to design a software application using UML. Therefore, object orientation is a must.

- Using Enterprise Architect is a must. If you using MAC, you can install EA using Wine.

ENTERPRISE ARCHITECT

Project Information (cont'd)

- You should prepare a project proposal including "PROJECT NAME", "GROUP MEMBERS", "CLIENT", and "PROJECT SUBJECT" with a brief explanation of your project and its features.

- You will prepare and submit a meeting report/s. Each meeting report should contain necessary information about your progress. Details can be found on Meeting Report Template in the attachments.

- Your design should include various diagrams. Details are given in UML Design Diagrams section.

UML Design Diagrams

- Your design should include the following diagrams:
  - Activity Diagrams
  - Use-Case Diagram
  - Context Diagram
  - Data Flow Diagrams
  - Entity-Relationship Diagram
  - Class Diagram
  - Sequence Diagrams
  - Communication Diagrams

Example Project Topics

- Android Task Monitoring
- Online Mobile Recharge Portal Project
- Vehicle Tracking Using Driver Mobile GPS Tracking
- Sentiment Analysis for Product Rating
- Fingerprint Based ATM System
- Image Encryption Using AES Algorithm
- Image Encryption Using Triple DES
- Extended AES with Custom Configurable Encryption

Example Project Topics (cont'd)

- E Commerce Product Rating Based On Customer Review Mining
- Canteen Automation System
- Engineering College Automation and Scheduling System
- Fingerprint Voting System Project
- Employee Hourly Attendance By Barcode Scan
- Weather Forecasting Using Data Mining
- Android Local Train Ticketing Project
- Android Patient Tracker

Grading and Deadlines

- Grade distribution:
  - Project Proposal
  - Meeting Report/s
  - Mid Progress Report
  - Final Report

- The impact of the project grade is 20%, keep this in mind!

- Additionally, you have to prepare Requirement Forms, and submit your Enterprise Architect file at the end of the semester along with your Final Report.

FIRST ASSIGNMENT

- Form a group of 4 or 5 people from your section only.

- Write your team members and project name (if you know) to this GoogleSheet below:

- Click here

FIRST ASSIGNMENT

- Prepare a project proposal including “PROJECT NAME”, “GROUP MEMBERS”, “CLIENT”, and “PROJECT SUBJECT” with a brief explanation of your project and its features.
- One of the team members should send your proposal to me via e-mail by writing SE 321/305 Project Proposal to the subject section.
- Please convert your file into .pdf before sending it to me.
- **Due date: 20th of October @23:59**
- Also print a copy of it and bring it to the LAB hours next week so that I can evaluate your Project.

Requirement Specification Forms

- You should fill 1 Requirement Specification Form for *each* requirement.

- At the end you need to have
  - at most 12 functional requirements,
  - and exactly 5 non-functional requirements.

REQUIREMENT SPECIFICATION FORM

Contact Information		
Project Name	Req. No	Req. Date
Beauty Book	2	09.11.2023
Requesting Firm or Organization		
By Serkan Erkek Kuaförü ve Güzellik Merkezi		
Contact Personel	Phone No	e-mail
Erhan Öztürk	0555 314 42 30	scanpolat@byserkan.com

Request Information
Request Description
The system should enable customers to view the availability of staffs and schedule appointments based on their preferences.
Justification
Allowing customers to see when staffs are available and book appointments online makes the process easier, saves time, and improves overall satisfaction. In addition to that customer have an opportunity to choose desired employee.
Effect if declined (Risk if the project request is declined)
Not having online scheduling may lead to slow bookings, unhappy customers, more work for staff, missed business chances, and falling behind competitors.
Request Alternatives (if exists)

Request Review Information		
Review Date	Reviewer Name and Signature	Reviewer Conclusion
09.11.2023	Utku Mert Çırakoğlu	☒ Approved ☐ Declined ☐ Postponed (D:)
09.11.2023	Yakup Atıcı	☒ Approved ☐ Declined ☐ Postponed (D:)
Suggestion Elaboration		

Request Final Review		
Review Date	Reviewer Name and Signature	Conclusion
09.11.2023	Sefa Kutay Berrak	☒ Approved ☐ Declined
Comments (if need)		

Functional and Non-Functional Req. Comparison

Aspect	Functional Requirements	Non-Functional Requirements
Definition	Describes what the system should do, i.e., specific functionality or tasks.	Describes how the system should perform, i.e., system attributes or quality.
Purpose	Focuses on the behavior and features of the system.	Focuses on the performance, usability, and other quality attributes.
Scope	Defines the actions and operations of the system.	Defines constraints or conditions under which the system must operate.
Examples	User authentication, data input/output, transaction processing.	Scalability, security, response time, reliability, maintainability.

SECOND ASSIGNMENT(Upcoming week)

- Please make an appointment with your clients. Conduct your interviews.
- Extract your functional and non-functional requirements.
- Prepare and write your requirement documents.

SECOND ASSIGNMENT (cont'd)

- Prepare Meeting Report #1, change its name to 'MeetingReport1_ProjectName_SectionNumber', convert to PDF, and send it to me.

- **Write your requirements to Meeting Report #1 attachment section. DO NOT SEND ME requirement specification forms please!!!**

- **Due date: TBA**

- Also print a copy of it and bring it to the LAB hours next week.

Requirement Analysis

Requirement Analysis Example 1

Read the following description of a requirement for FoodCo, and decide which parts of it are functional requirements and which are non-functional requirements.

The allocation of staff to production lines should be mostly automated. A process will be run once a week to carry out the allocation based on the skills and experience of operatives. Details of holidays and sick leave will also be taken into account. A first draft Allocation List will be printed off by 12:00 noon on Friday for the following week. Only staff in Production Planning will be able to amend the automatic allocation to fine-tune the list. Once the amendments have been made, the final Allocation List must be printed out by 5.00 pm. The system must be able to handle allocation of 700 operatives at present, and should be capable of expansion to handle double that number.

Requirement Analysis Example

Functional requirements are:

- the need for a process to be run that allocates staff to lines based on their skills and experience, holidays and sick leave;
- printing out an allocation list;
- amending the allocation list.

Examples of non-functional requirements are:

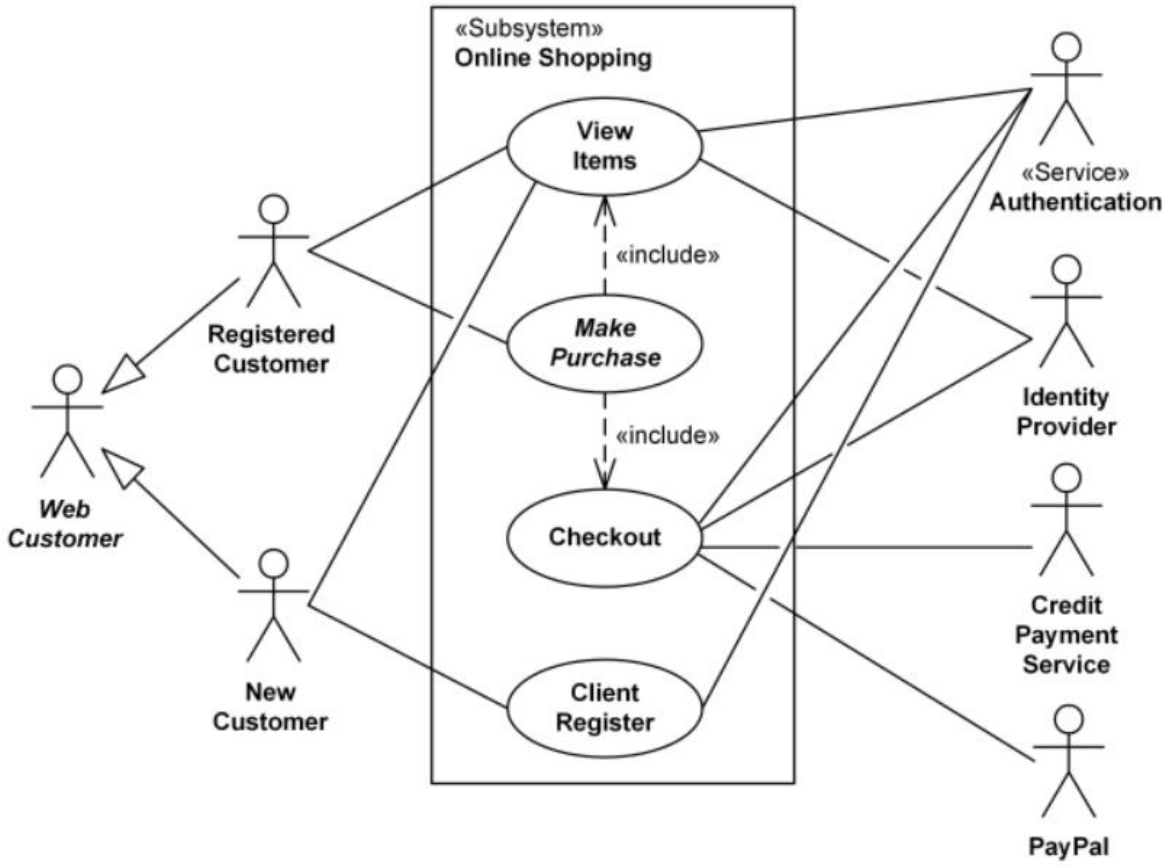
- printing the allocation list by 12:00 noon;
- the need to handle 700 operatives' details

A Use Case Example

- Web Customer actor uses some web site to make purchases online. Top level use cases are View Items, Make Purchase and Client Register. View Items use case could be used by customer as top level use case if customer only wants to find and see some products. This use case could also be used as a part of Make Purchase use case. Client Register use case allows customer to register on the web site, for example to get some coupons or be invited to private sales. Note, that Checkout use case is included use case not available by itself - checkout is part of making purchase.

- Except for the Web Customer actor there are several other actors which will be described below with detailed use cases.

Use Case Diagram of Given Scenario



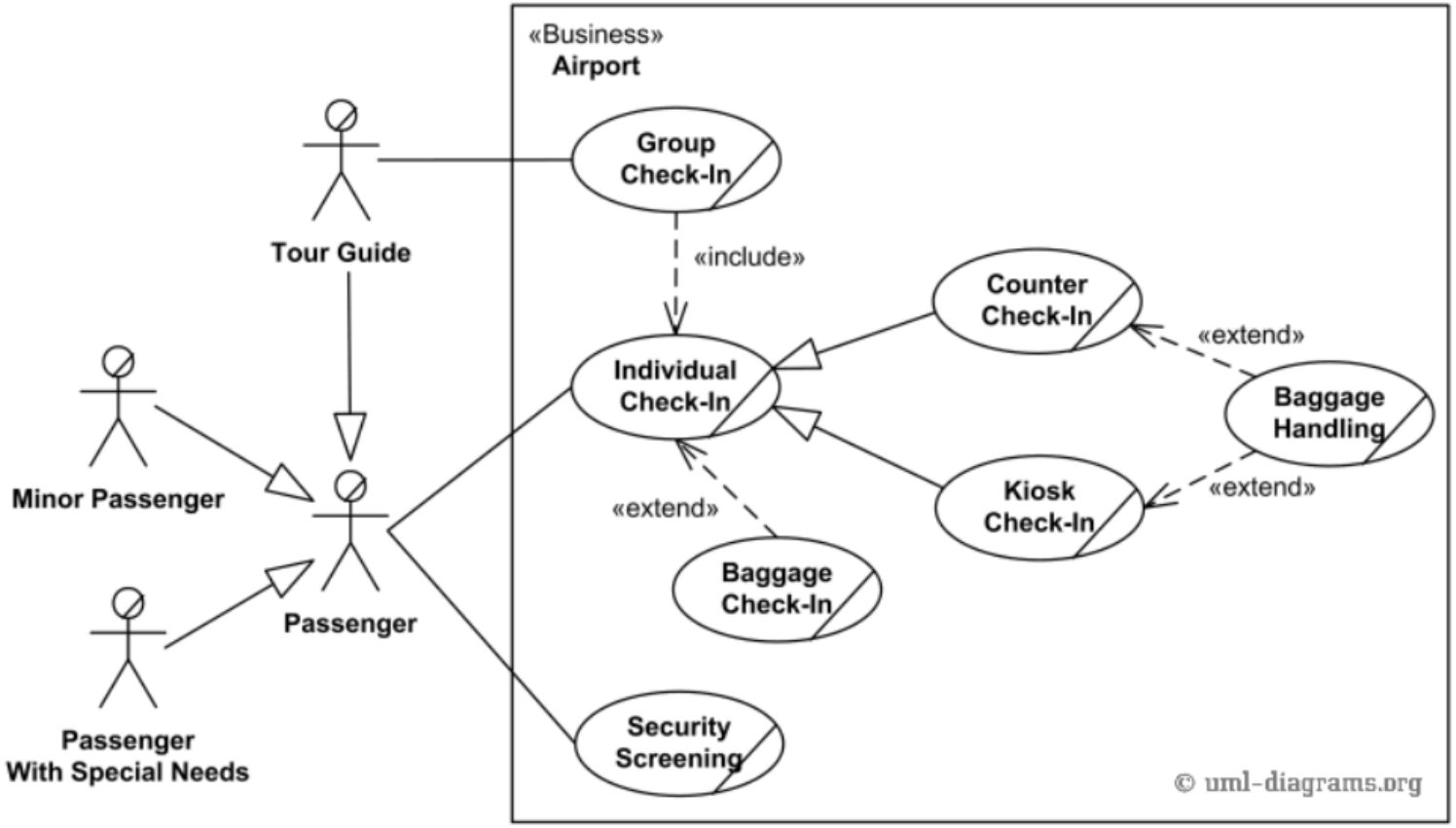
Another Use Case example

- This is an example of a business use case diagram which is usually created during Business Modeling and is rendered here in Rational Unified Process (RUP) notation.

- Business actors are Passenger, Tour Guide, Minor (Child), Passenger with Special Needs (e.g. with disabilities), all playing external roles in relation to airport business.

- Business use cases are Individual Check-In, Group Check-In (for groups of tourists), Security Screening, etc. - representing business functions or processes taking place in airport and serving the needs of passengers.

Use Case Diagram



Business use cases Baggage Check-in and Baggage Handling extend Check-In use cases, because passenger might have no luggage, so baggage check-in and handling are optional.